

Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	Rising Water
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	React.js	Enables responsive design and component reusability for a seamless user experience.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Node.js (Express)	Provides fast processing and leverages existing team expertise for robust server-side logic.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MangoDB	Chosen based on whether the project needs to handle unstructured data or requires strong relational mapping.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	ASW(Amazon Web Service)	Ensures easy scalability and cost-effective hosting for deployment and scaling needs.
5	Version Control & CI/CD	Placeholder	Facilitates seamless collaboration and automated testing for continuous integration and delivery.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	APlaceholder where youneed to list the actual tools or APIs your project uses	Adds specific features such as payment processing, messaging, or mapping to enhance project capabilities.